that is
$$\begin{array}{rl} y = & x^2 + x - 2 \\ 0 = & x^2 + x - 2 \\ \hline y = & 0 \end{array} \quad (-)$$

The equation $y = 0$ represents the X axis.

Step 3: Mark the point of intersection of the curve $y = x^2 + x - 2$ with the X axis. That is $(-2,0)$ and $(1,0)$

Step 4: The x coordinates of the respective points form the solution set $\{-2,1\}$ for $x^2 + x - 2 = 0$

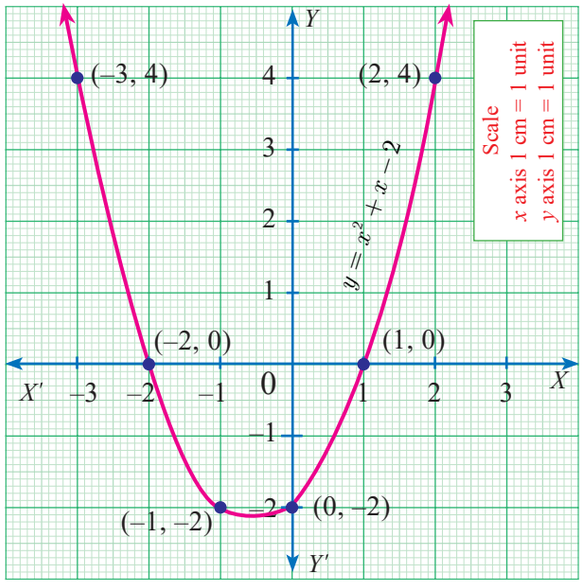


Fig. 3.24

Example 3.55 Draw the graph of $y = x^2 - 4x + 3$ and use it to solve $x^2 - 6x + 9 = 0$

Solution

Step 1: Draw the graph of $y = x^2 - 4x + 3$ by preparing the table of values as below

x	–2	–1	0	1	2	3	4
y	15	8	3	0	–1	0	3

Step 2: To solve $x^2 - 6x + 9 = 0$, subtract $x^2 - 6x + 9 = 0$ from $y = x^2 - 4x + 3$

that is
$$\begin{array}{rl} y = & x^2 - 4x + 3 \\ 0 = & x^2 - 6x + 9 \\ \hline y = & 2x - 6 \end{array} \quad (-)$$

The equation $y = 2x - 6$ represent a straight line. Draw the graph of $y = 2x - 6$ forming the table of values as below.

x	0	1	2	3	4	5
y	–6	–4	–2	0	2	4

The line $y = 2x - 6$ intersect $y = x^2 - 4x + 3$ only at one point.

Step 3: Mark the point of intersection of the curve $y = x^2 - 4x + 3$ and $y = 2x - 6$ that is $(3,0)$.

Therefore, the x coordinate 3 is the only solution for the equation $x^2 - 6x + 9 = 0$.

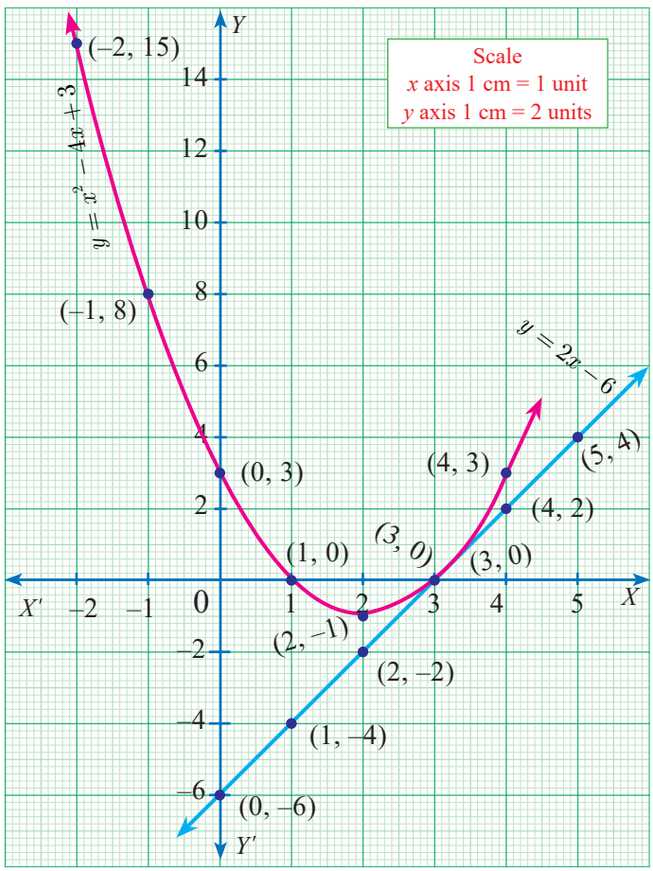


Fig. 3.25

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